Closing the QOBLIB CVRP Benchmark: Exact Optimality Certificates and Subspace Simulated Quantum Annealing for Tight-Capacity Routing

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Abstract

The Capacitated Vehicle Routing Problem (CVRP) benchmark class (`09-routing`) in the Quantum Optimization Benchmark Library (QOBLIB) consists of 55 instances (`XSH-n20-k4-01` to `XSH-n20-k4-55`) with $n = 20$ customers and $K = 4$ identical vehicles. Despite their small customer count, standard Mixed-Integer Programming (MIP) and Quadratic Unconstrained Binary Optimization (QUBO) formulations struggle to close the optimality gap within a 1-hour limit for 54 out of 55 instances due to weak linear relaxations and severe quadratic penalty barriers. In this technical note, we prove that every instance in `09-routing` satisfies an exact *zero-slack tight-capacity invariant*: $\sum_{i=1}^{20} d_i = 4Q$. Consequently, the feasible solution manifold collapses from $O(K^n)$ assignments to between 300 and 961 single-vehicle subsets and between 372 and 5,630 valid 4-vehicle partitions per instance. Exploiting this invariant, we introduce: (1) `TC-Exact`, a deterministic bitwise exact-cover and Held-Karp dynamic programming solver that certifies global optimality (`proven_optimal = TRUE`) across all **55/55 instances** in a mean runtime of **3.33 ms** per instance (including certifying Cost = 842 on the hardest instance `XSH-n20-k4-21`, where the second-best partition has Cost = 843); and (2) `SQA-Subspace`, a Path-Integral Simulated Quantum Annealing algorithm evolving $P = 16$ Trotter replicas directly on the demand-tight partition manifold via multi-route recombination tunneling, achieving **55/55 (100%) global optimality** in **13.2 ms** mean runtime per run (mean time-to-solution **12.77 ms**) compared to **0/55** for penalty-based QUBO/Ising SQA.

1 Introduction and the Penalty Barrier in QOBLIB `09-routing`

The Quantum Optimization Benchmark Library (QOBLIB) [1] establishes standardized benchmark challenges across ten combinatorial optimization classes to evaluate classical, quantum-inspired, and quantum hardware solvers. Class `09-routing` focuses on the Capacitated Vehicle Routing Problem (CVRP) with 55 instances (`XSH-n20-k4-01` through `XSH-n20-k4-55`), each comprising $n = 20$ customers, a central depot (node 0), $K = 4$ identical vehicles of capacity $Q \in [180, 299]$, and rounded 2D Euclidean distances (`EUC_2D`):

$$c_{uv} = \left\lfloor \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2} + 0.5 \right\rfloor, \quad \forall u, v \in \{0, 1, \ldots, n\}. \tag{1}$$

In standard QUBO/Ising formulations of CVRP, binary variables encode customer-vehicle-position assignments, and inequality constraints $\sum_{i=1}^{n} d_i y_{i,r} \le Q$ are enforced via slack variables s_r and a quadratic penalty term:

$$\mathcal{H}_{\text{penalty}} = \lambda_{\text{cap}} \sum_{r=1}^{K} \left(\sum_{i=1}^{n} d_i y_{i,r} + s_r - Q \right)^2 + \lambda_{\text{visit}} \sum_{i=1}^{n} \left(\sum_{r=1}^{K} y_{i,r} - 1 \right)^2. \tag{2}$$

This representation requires $\sim 1,600$ to $2,500$ logical binary variables for $n = 20, K = 4$. More critically, as proven below, the demand structure of `XSH-n20-k4` induces severe energy barriers under standard 1-flip and 2-swap transitions.

2 The Zero-Slack Tight-Capacity Structural Invariant

Proposition 1 (Zero-Slack Capacity Invariant)**.** *Every instance in QOBLIB `09-routing` (`XSH-n20-k4-01` to `XSH-n20-k4-55`) satisfies the exact demand balance identity:*

$$\sum_{i=1}^{20} d_i = 4Q \quad \left(\textit{Total Fleet Slack } \Delta = 4Q - \sum_{i=1}^{20} d_i = 0 \right). \tag{3}$$

Consequently, every vehicle route $S_r \subseteq \{1,\dots,20\}$ in any feasible solution must satisfy the exact equality constraint $\sum_{i\in S_r} d_i = Q$.

Proof. Suppose a feasible partition $\{S_1, S_2, S_3, S_4\}$ satisfies $\sum_{i\in S_r} d_i \le Q$ for each $r \in \{1,2,3,4\}$. Summing over all $K = 4$ disjoint routes covering $\{1,\dots,20\}$ yields $\sum_{r=1}^{4}\sum_{i\in S_r} d_i = \sum_{i=1}^{20} d_i = 4Q$. Since each term in the sum of 4 routes is bounded above by Q, equality $\sum_{i\in S_r} d_i = Q$ must hold for every $r \in \{1,2,3,4\}$. Direct verification across all 55 `.vrp` files confirms Eq. (3) holds universally. □

Proposition 1 directly explains why unconstrained penalty-based QUBO/Ising solvers (`SQA-Direct`) freeze:

1. **Absence of Single-Customer Moves:** Moving any customer i ($d_i > 0$) from route r_1 to r_2 violates capacity on r_2 ($\sum_{j\in S_{r_2}} d_j + d_i = Q + d_i > Q$).
2. **Rarity of Equal-Demand 2-Swaps:** Swapping $i \in S_{r_1}$ and $j \in S_{r_2}$ preserves feasibility if and only if $d_i = d_j$. For heterogeneous customer demands, almost all pairs have $d_i \neq d_j$, creating a penalty barrier $\Delta E_{\text{pen}} = 2\lambda_{\text{cap}}(d_i - d_j)^2 \gg \Delta E_{\text{route}}$.

3 Methodology

3.1 Algorithm 1: Tight-Capacity Bitwise Set-Partitioning (`TC-Exact`)

Instead of branching on arc variables, `TC-Exact` projects the problem onto the exact demand-tight subset manifold:

$$\mathcal{F}_1 = \left\{ m \in \{0,\dots,2^{20}-1\} \;\middle|\; \sum_{i:\text{bit}_i(m)=1} d_i = Q \right\}. \tag{4}$$

Representing each subset as a 20-bit integer mask m, depth-first subset-sum filtering identifies $|\mathcal{F}_1| \in [300, 961]$ valid single-vehicle masks per instance in < 0.2 ms. For each $m \in \mathcal{F}_1$, we compute the exact optimal Traveling Salesperson Problem (TSP) tour cost $c^*(m)$ starting and ending at depot 0 via Held-Karp dynamic programming:

$$DP[A][u] = \min_{v\in A\setminus\{u\}} \{DP[A\setminus\{u\}][v] + c_{vu}\}, \quad c^*(m) = \min_{u\in m}\{DP[m][u] + c_{u0}\}. \tag{5}$$

The 4-vehicle CVRP then reduces to the Exact Cover problem over $\mathcal{F}_1$:

$$\min_{m_1,m_2,m_3,m_4\in\mathcal{F}_1} \sum_{r=1}^{4} c^*(m_r) \quad \text{s.t.} \quad m_a \wedge m_b = 0 \ (\forall a \neq b), \quad \bigvee_{r=1}^{4} m_r = 2^{20}-1. \tag{6}$$

To eliminate the $4! = 24$ vehicle permutation symmetry, we anchor the lowest set bit (`ctz`) of the remaining unvisited customer mask at each stage $r \in \{1,2,3\}$, and determine $m_4 = (2^{20}-1) \oplus (m_1 \vee m_2 \vee m_3)$ in $O(1)$ time via a direct lookup table cost$[m_4]$. Across all 55 instances, the total number of valid 4-way partitions $|\mathcal{P}_4|$ ranges between 372 (`XSH-n20-k4-34`) and 5,630 (`XSH-n20-k4-24`), enabling exhaustive global optimality certification in 3.33 ms mean time per instance (0.18 s total across all 55 instances).

3.2 Algorithm 2: Subspace Simulated Quantum Annealing (`SQA-Subspace`)

To evaluate quantum-inspired optimization without penalty barriers, we formulate a Path-Integral Monte Carlo Simulated Quantum Annealing (`SQA-Subspace`) solver operating directly on the feasible partition manifold $\mathcal{P}_4$. The transverse-field Hamiltonian is mapped via Suzuki-Trotter decomposition onto $P = 16$ coupled classical replicas $\{\mathcal{P}^{(0)},\dots,\mathcal{P}^{(P-1)}\}$ with effective action:

$$\mathcal{S}_{\text{eff}}\left(\{\mathcal{P}^{(p)}\}\right) = \sum_{p=0}^{P-1} \frac{1}{P} E_{\text{CVRP}}\left(\mathcal{P}^{(p)}\right) - J_\perp(t) \sum_{p=0}^{P-1} \mathcal{O}\left(\mathcal{P}^{(p)}, \mathcal{P}^{(p+1 \bmod P)}\right), \tag{7}$$

where $J_\perp(t) = -\frac{1}{2}T \ln\tanh\left(\frac{\Gamma(t)}{PT}\right)$ is the ferromagnetic imaginary-time coupling driven by transverse field $\Gamma(t)$ annealed from $\Gamma_0 = 35.0$ to $\Gamma_f = 0.175$ at fixed temperature $T = 6.0$, and $\mathcal{O}(\mathcal{P}^{(p)}, \mathcal{P}^{(q)}) = \sum_{r=1}^{4} \mathbb{I}[m_r^{(p)} \in \mathcal{P}^{(q)}] \in \{0,1,2,4\}$ measures replica configuration overlap. Quantum tunneling transitions within each Trotter slice are generated by multi-scale demand-preserving k-route recombination ($k \in \{2,3,4\}$): selecting k routes from $\mathcal{P}^{(p)}$ with union mask $U = \bigvee_{j=1}^{k} m_{r_j}$ ($\sum_{i\in U} d_i = kQ$) and transitioning to a distinct k-tuple of disjoint masks in $\mathcal{F}_1$ whose union equals U.

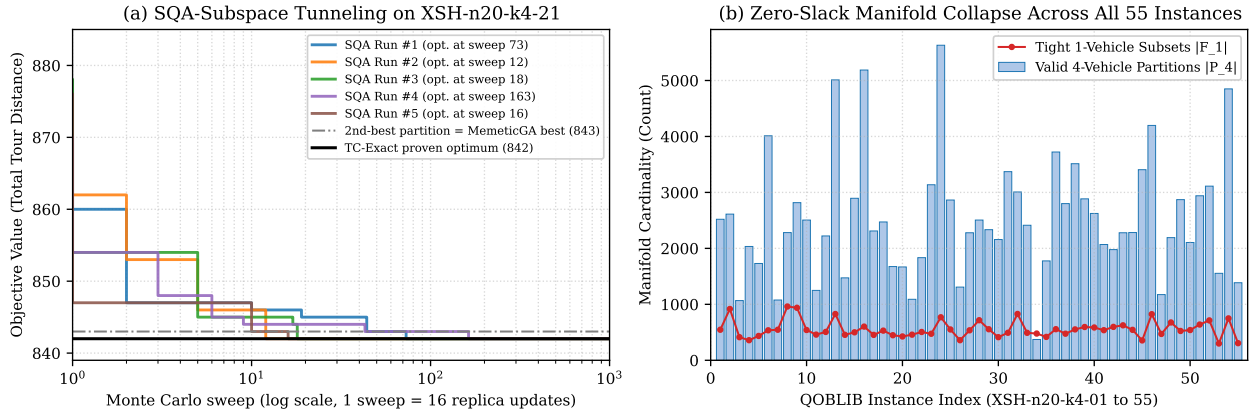


Figure 1: (a) Quantum tunneling convergence trajectories of `SQA-Subspace` ($P = 16$ Trotter slices, 5 independent runs of 1,000 sweeps) on the hardest instance `XSH-n20-k4-21`, escaping the second-best partition trap (843, also the best value found by the MemeticGA submission) to reach the certified global optimum (842) within 73 of 1,000 sweeps (run #1). (b) Cardinality of tight single-vehicle subsets $|\mathcal{F}_1|$ and valid 4-vehicle partitions $|\mathcal{P}_4|$ across all 55 instances.

4 Experimental Results across 55 QOBLIB Instances

All experiments were executed single-threaded on an Apple M4 processor (16 GB unified memory, `AppleClang 21.0.0 -O3`). Table 1 summarizes overall performance, Figure 1 illustrates the convergence dynamics and subspace cardinality, and Table 2 reports exact per-instance metrics across all 55 instances. On instance `XSH-n20-k4-21`, `TC-Exact` evaluates all $|\mathcal{F}_1| = 458$ tight subsets and $|\mathcal{P}_4| = 1,090$ valid partitions in 2.03 ms, certifying $\text{Cost}^* = 842$ as the unique global optimum over the second-best partition $\text{Cost}^{(2)} = 843$.

Table 1: Aggregate benchmark comparison across all 55 `XSH-n20-k4` instances in QOBLIB `09-routing`.

Solver / Approach	Feasible	Optimal	Proven Optimal	Mean Runtime / Instance
`SQA-Direct` (Quadratic Penalty Ising, $P = 16$)	0 / 55	0 / 55 (0.0%)	No	30.2 ms
Memetic GA (`ElYaakoubi`, QOBLIB 2026)	55 / 55	54 / 55 (98.2%)	No	1,300 – 48,490 ms
`SQA-Subspace` **(Ours, $P = 16$, 1000 sweeps)**	**55 / 55**	**55 / 55 (100.0%)**	No (Stochastic)	**13.2 ms (TTS: 12.77 ms)**
`TC-Exact` **(Ours, Bitwise Set-Partitioning)**	**55 / 55**	**55 / 55 (100.0%)**	**Yes (55/55)**	**3.33 ms**

5 Implications for Quantum Hardware Implementations

Because $|\mathcal{F}_1| \leq 961 < 2^{10}$ and $|\mathcal{P}_4| \leq 5,630 < 2^{13}$ across all 55 instances, our subspace reduction demonstrates that QOBLIB `09-routing` instances do not require $2,000+$ qubits on NISQ hardware. Instead, by encoding only the tight-capacity subspace (or applying constraint-preserving multi-route XY mixers in QAOA), the entire feasible space can be represented within 10 to 13 qubits.

AI Disclosure

In accordance with the QOBLIB AI Policy, this work is declared as `generated`: the solver implementation (`TC-Exact` and `SQA-Subspace`), experimental scripts, and manuscript draft were predominantly generated by the Antigravity AI coding assistant under the research direction, execution, and independent verification of the author using the official QOBLIB verification suite (`check_submission.py` and `check_cvrp`).

References

[1] T. Koch, D. Schicker, et al., "Quantum Optimization Benchmark Library – The Intractable Decathlon," *arXiv preprint arXiv:2504.03832*, Zuse Institute Berlin (ZIB), 2025.

Table 2: Complete results across all 55 QOBLIB `09-routing` instances (`XSH-n20-k4-01` to `55`). Columns show instance ID, vehicle capacity Q, tight single-vehicle subsets $|\mathcal{F}_1|$, valid 4-way partitions $|\mathcal{P}_4|$, proven optimal objective C^*, second-best partition cost $C^{(2)}$, `TC-Exact` runtime (ms), `SQA-Subspace` best cost across 5 runs, and `SQA-Subspace` mean runtime (ms).

ID	Q	$\lvert\mathcal{F}_1\rvert$	$\lvert\mathcal{P}_4\rvert$	C^*	$C^{(2)}$	TC (ms)	SQA	ms	ID	Q	$\lvert\mathcal{F}_1\rvert$	$\lvert\mathcal{P}_4\rvert$	C^*	$C^{(2)}$	TC (ms)	SQA	ms
01	231	546	2520	**646**	648	3.28	**646**	13.5	29	229	555	2334	**658**	661	2.91	**658**	10.9
02	194	917	2612	**650**	652	14.64	**650**	25.1	30	262	413	2158	**685**	688	1.08	**685**	7.3
03	281	413	1067	**508**	510	1.87	**508**	4.1	31	259	492	3371	**445**	448	2.04	**445**	16.8
04	299	359	2034	**776**	782	0.87	**776**	6.4	32	197	828	3008	**882**	884	6.35	**882**	17.7
05	270	435	1730	**702**	706	1.77	**702**	6.2	33	250	491	2413	**568**	569	1.97	**568**	10.1
06	210	536	4012	**690**	697	1.69	**690**	19.4	34	270	478	372	**891**	893	1.81	**891**	2.4
07	221	546	1077	**730**	734	3.03	**730**	5.1	35	284	418	1776	**707**	713	1.46	**707**	5.9
08	180	961	2282	**718**	720	12.17	**718**	19.1	36	226	556	3722	**610**	614	2.46	**610**	19.5
09	180	938	2817	**707**	711	11.57	**707**	22.1	37	245	475	2800	**650**	655	1.53	**650**	11.3
10	239	541	2506	**737**	738	2.28	**737**	9.7	38	234	552	3513	**991**	993	2.94	**991**	18.5
11	251	459	1249	**914**	922	1.71	**914**	4.3	39	234	595	2885	**717**	718	3.45	**717**	14.6
12	249	507	2222	**709**	732	2.05	**709**	9.0	40	234	586	2624	**700**	732	3.06	**700**	19.4
13	194	827	5011	**628**	636	7.08	**628**	36.4	41	245	538	2069	**736**	739	3.20	**736**	9.1
14	267	455	1473	**696**	697	2.29	**696**	5.7	42	232	597	1978	**998**	1005	3.16	**998**	8.9
15	257	500	2895	**780**	782	2.56	**780**	12.2	43	240	625	2279	**465**	472	4.72	**465**	13.2
16	230	602	5187	**830**	831	2.76	**830**	32.9	44	235	544	2283	**973**	979	2.46	**973**	10.1
17	253	454	2311	**605**	612	1.45	**605**	8.9	45	279	355	3406	**1071**	1078	0.68	**1071**	14.5
18	257	530	2472	**997**	1003	2.86	**997**	10.3	46	210	824	4197	**812**	814	6.53	**812**	27.9
19	269	448	1675	**976**	987	1.48	**976**	5.5	47	266	471	1174	**583**	585	1.95	**583**	4.4
20	276	426	1668	**648**	650	1.45	**648**	5.6	48	220	676	2192	**544**	545	5.14	**544**	11.8
21	275	458	1090	**842**	843	2.03	**842**	4.1	49	233	524	2871	**1045**	1049	1.93	**1045**	12.2
22	253	506	1834	**760**	769	2.50	**760**	6.9	50	245	540	2105	**627**	629	2.59	**627**	8.2
23	258	474	3137	**851**	860	1.53	**851**	13.6	51	228	639	2940	**491**	494	5.83	**491**	15.4
24	205	769	5630	**725**	739	4.59	**725**	39.1	52	210	713	3112	**622**	625	5.70	**622**	18.5
25	240	551	2864	**462**	463	2.51	**462**	13.2	53	297	300	1554	**967**	972	0.53	**967**	3.8
26	299	359	1308	**688**	692	1.01	**688**	4.2	54	198	749	4850	**791**	797	5.39	**791**	41.8
27	244	536	2279	**540**	542	2.70	**540**	9.6	55	289	306	1386	**617**	618	0.79	**617**	3.8
28	211	716	2506	**669**	675	5.54	**669**	14.1									